

Project Proposal

WikiLLM-Based Intelligent Knowledge Management & Question Answering System

Submitted By

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Position

AI Intern – Technical Assignment

1. Project Overview

This project proposes the development of an intelligent document understanding and question-answering system using the **WikiLLM architecture**. Unlike traditional Retrieval-Augmented Generation (RAG) systems that rely primarily on vector databases and semantic similarity search, this system transforms uploaded documents into a structured, interconnected knowledge base before enabling question answering.

The system will automatically analyze uploaded documents, extract entities, concepts, relationships, and factual information, and organize them into a wiki-like knowledge repository. Users will then be able to ask natural language questions, and the system will generate accurate answers based on the structured knowledge.

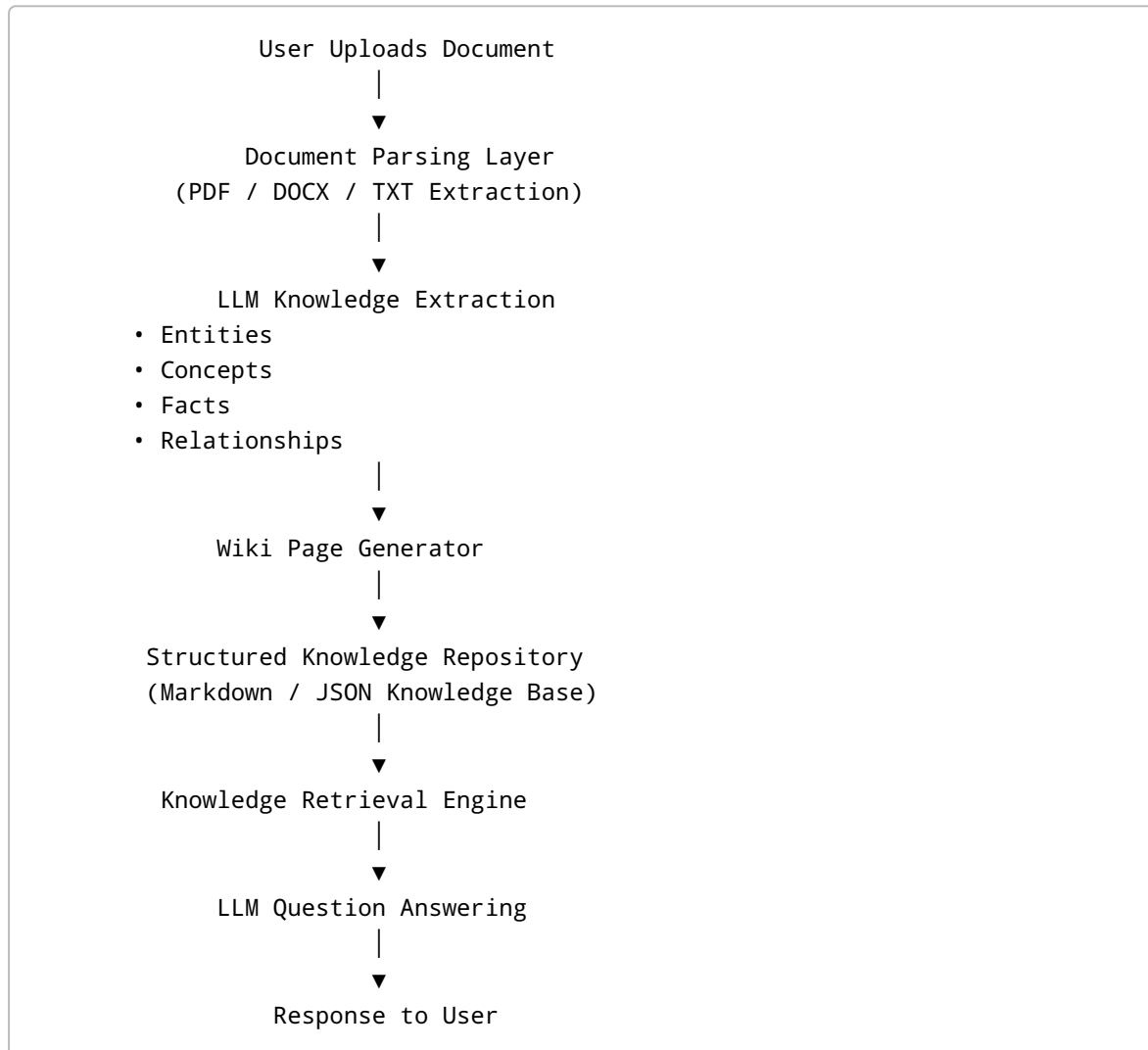
The objective is to demonstrate how Large Language Models can be used not only for retrieval but also for knowledge organization and intelligent reasoning.

2. Objectives

The primary objectives of this project are:

- Develop a document ingestion pipeline.
 - Extract structured knowledge from uploaded documents.
 - Generate interconnected Wiki pages automatically.
 - Store knowledge in an organized, reusable format.
 - Enable natural language question answering over the generated knowledge.
 - Build a scalable architecture that can support multiple documents in the future.
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3. Proposed System Architecture



4. Proposed Workflow

Step 1 – Document Upload

Users upload one or more documents (PDF, DOCX, or TXT).

Step 2 – Document Parsing

The system extracts readable text from the uploaded document while preserving document structure where possible.

Step 3 – Knowledge Extraction

Using an LLM, the system identifies:

- Important entities
- Key concepts
- Definitions
- Facts
- Relationships between entities
- Important summaries

The extracted knowledge will be converted into structured JSON.

Step 4 – Wiki Generation

The extracted knowledge is automatically converted into wiki-style pages.

Example:

```
Tesla.md

Overview

Founder

CEO

Products

Related Topics

[[Electric Vehicles]]

[[Battery Technology]]
```

Each page will contain references to related topics to create an interconnected knowledge network.

Step 5 – Knowledge Storage

Generated wiki pages will be stored in a structured repository.

Possible formats include:

- Markdown
 - JSON
 - SQLite metadata
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Step 6 – Question Answering

When a user submits a question:

1. Relevant wiki pages are identified.
 2. Related linked pages are collected.
 3. Context is sent to the LLM.
 4. The LLM generates the final answer.
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5. Key Features

- Document Upload
 - Automatic Knowledge Extraction
 - Entity Recognition
 - Relationship Detection
 - Wiki Page Generation
 - Linked Knowledge Pages
 - Natural Language Question Answering
 - Multi-document Support
 - Persistent Knowledge Base
 - Easily Extendable Architecture
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6. Proposed Technology Stack

Component	Technology
Programming Language	Python
Backend	FastAPI
LLM Framework	LangGraph / LangChain
LLM	OpenAI GPT / Local LLM
Document Parsing	PyMuPDF
Knowledge Storage	Markdown + JSON
Metadata Storage	SQLite
Optional Semantic Search	ChromaDB / FAISS
Version Control	Git

7. Deliverables

The completed project will include:

- Source Code
 - Project Documentation
 - System Architecture Diagram
 - Sample Knowledge Base
 - Demonstration with Sample Documents
 - Question Answering Demo
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8. Development Plan

Day 1

- Research WikiLLM architecture
- Design overall system
- Implement document parsing
- Implement knowledge extraction prompts

Day 2

- Generate wiki pages
- Implement structured knowledge storage
- Develop question-answering workflow

Day 3

- Testing
 - Performance improvements
 - Documentation
 - Final demonstration
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9. Future Enhancements

Future improvements may include:

- Multi-user support
- Knowledge graph visualization
- Incremental document updates
- Contradiction detection
- Automatic knowledge merging
- Hybrid search (Wiki + Vector Search)
- Web interface
- Role-based access control
- Source citation and traceability

10. Expected Outcome

The final system will demonstrate an intelligent document understanding workflow that goes beyond traditional vector-based RAG systems. Instead of retrieving isolated document chunks, the system will organize information into a structured wiki-like knowledge base, enabling more accurate, explainable, and reusable knowledge retrieval for question answering.

This architecture showcases the ability of Large Language Models to perform knowledge extraction, organization, and reasoning while maintaining an extensible foundation for future enhancements.